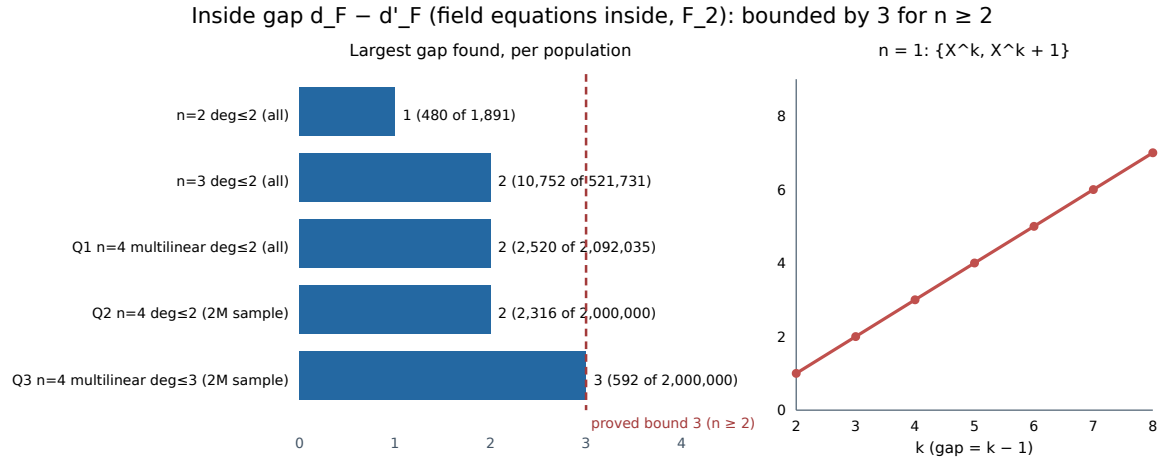
The inside fake-to-true gap: bounded by 3 for n ≥ 2, unbounded for n = 1 (2026-11-02)

Verdict. The question `GAP-FAKE-GAP-INSIDE-BOUNDED` is settled. - **n ≥ 2 variables:** with the field equations inside the generating set, `d_F − d'_F ≤ 3` for every system over `F_2`, and 3 is attained (n = 4). - **One variable:** the gap is unbounded.

The bound comes from a fall built from the field equations alone, so it also shows that, under the equality reading, inside-placement `d_F` never exceeds 4. All results are exact. No cost claim.

Protocol: [PROTOCOL.md](), frozen before computing, with Amendments 1–2. Definitions: Nagao ePrint 2015/984 Definitions 5 and 6, equality reading, over `F_2`.



For n ≥ 2: Y²(X²+X) + X²(Y²+Y) = XY² + X²Y is a fall at 4 for every system, so d_F(inside) ≤ 4 and gap ≤ 3.

inside gap

The search

Instrument: `inside_gap_n4.rs`. Lines 1..384 are a byte-for-byte copy of the `blindspots_19_checks.rs` fall computations. The driver adds threads, seeded sampling, and the inside gap only. Control: the n = 3 witness embedded in n = 4 gives gap 2 (PASS).

population	systems	max gap	systems at the max	undetermined
n = 2, all pairs deg ≤ 2	1,891	1	480	0
n = 3, all pairs deg ≤ 2	521,731	2	10,752	0
n = 4, all pairs multilinear deg ≤ 2 (Q1)	2,092,035	2	2,520	0
n = 4, pairs deg ≤ 2, seeded sample (Q2)	2,000,000	2	2,316	0
n = 4, pairs multilinear deg ≤ 3, seeded sample (Q3)	2,000,000	**3**	592	0

The n = 3 row reproduces the earlier census of `research/lean_attacks_20261102` exactly.

Gap-3 witness. `{XYW + XY + W, XZW + XW + 1}` has `d_F(inside) = 4` and `d'_F = 1`. It is confirmed by `independent_falls.py`, an exact linear-algebra implementation in Python that shares no code with the Rust instrument.

Why it stops at 3

The witnesses share a shape, `F(A, B, v) = {A v + 1, B v + B + 1}`: `- (1 + v) f_1 + v f_2` reduces to `1`, so `d'_F = 1`; `- v f_2 + B (v² + v) = v` is a true fall (inside) at `deg B + 2`.

That suggested a gap growing with `deg A, deg B`. Evaluating the family (Amendment 1) refuted it: `d_F(inside)` follows `min(deg A, deg B) + 2` up to 4, then stays at 4 for every larger `k`. Both implementations agree (`results/family_rust.txt`).

The independent implementation's degree-4 fall uses only the field equations:

`Y²(X² + X) + X²(Y² + Y) = XY² + X²Y`

Both products have degree 4, and the sum has degree 3 and is nonzero.

Theorem (n ≥ 2). For every `F ⊆ F_2[X_1, …, X_n]` with `n ≥ 2`, `d_F(F ∪ S_fe) ≤ 4`, and therefore `d_F(inside) − d'_F ≤ 3`.

Proof. 1. Put multiplier `X_2²` on `X_1² + X_1`, multiplier `X_1²` on `X_2² + X_2`, and 0 on every other generator. 2. The two products have degree 4 and the sum `X_1 X_2² + X_1² X_2` has degree 3 and is nonzero, so this is a fall at 4 and `d_F ≤ 4`. 3. `d'_F ≥ 1` always, since a fall needs a sum of lower degree than `d`. ∎

Checks: the field equations alone give `d_F = 4` for n = 2..8 (Rust) and n = 2..5 (Python).

Tightness. The gap-3 witness above.

One variable. With n = 1 there is no second field equation. `{X^k, X^k + 1}`: `- X + (X + 1) = 1` is a fake fall at 1; `- X^k + (X^k + 1) = 1` is a true fall at k; - below k only `X² + X` is admissible, and one generator never falls.

So the gap is `k − 1`, unbounded; checked k = 2..8 by both implementations.

What it means

- **Inside-placement `d_F` is degenerate.** Under the equality reading it is at most 4 for every system in two or more variables, because of falls among the field equations themselves. These are the "trivial degree falls" that Ding–Hodges and Petit–Quisquater exclude from their first fall degree; the definition encoded in `papers/ic-negative-results` does not exclude them.
- **Consequences for the inside placement:**
  - Nagao's Lemma 4 can fail there only when `d'_F < 4`.
  - No estimate can take its degree from inside-placement `d_F` without first removing trivial falls.
- **Outside the gap is unbounded** (`W_k`, previous round). Inside it is bounded, but only because the quantity itself is capped.

Formal status

- `papers/ic-negative-results/lean/IcNegative/InsideGap.lean` states three things in core Lean with `decide +kernel`: the field-equation fall at 4 in two variables; no fall below 4 from the field equations alone; and the same fall inside witness A's system. The repository proof gate passed it (run 41484047855: target PASS, 44 theorems, 4 of them new, 0 trusted).
- The step "zero multipliers on further generators leave the fall intact" is argued, not formalised.

- The four-variable witnesses are native only (two implementations), since the Lean encoding covers two variables.

Reproduce

```
        D=research/inside_gap_n4_20261102
        rustc --edition 2021 -C opt-level=3 -o /tmp/ig4 $D/inside_gap_n4.rs
        /tmp/ig4 control; /tmp/ig4 series; /tmp/ig4 q1; /tmp/ig4 q2 2000000; /tmp/ig4 q3
2000000   # ~7 s, ~70 s, ~81 s, ~108 s (4 threads)
        rustc --edition 2021 -C opt-level=3 -o /tmp/fam8 $D/family_n8.rs && /tmp/fam8
        python3 -I $D/independent_falls.py --selftest
        python3 -I $D/independent_falls.py "XYW+XY+W; XZW+XW+1" 4 8
        python3 -I $D/plot_inside_gap.py
```